COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 212 - Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	Aug. 21, 2026	**DATE SUBMITTED:**	Aug. 25, 2026
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ACTIVITY NO. 4 **Ansible Basics**

1. PROCEDURE OUTPUT

Task 1: Run elevated ad hoc commands

1. Locally, we use the command sudo apt update when we want to download package information from all configured resources. The sources are often defined in /etc/apt/sources.list file and other files located in /etc/apt/sources.list.d/ directory. So, when you run the update command, it downloads the package information from the Internet. It is useful to get info on an updated version of packages or their dependencies. We can only run an apt update command in a remote machine. Issue the following command:
ansible all -m apt -a update_cache=true

What is the result of the command? Is it successful?

- The command failed on all of the servers. The reason why it failed because it did not have an administrator privileges which causes the permission error.

```
charles-vence-espinas@workstation:~/ansible-lab$ ansible all -m apt -a update_cache=true
[WARNING]: Host 'centOS' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of
another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-co
re/2.21/reference_appendices/interpreter_discovery.html for more information.
[WARNING]: Updating cache and auto-installing missing dependency: python3-apt
[ERROR]: Task failed: Module failed: Error executing command: [Errno 2] No such file or directory: b'update'
Origin: <adhoc 'apt' task>

{'action': 'apt', 'args': {'update_cache': 'true'}, 'timeout': 0, 'async_val': 0, 'poll': 15}

centOS | FAILED! => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.9"
    },
    "changed": false,
    "cmd": "update",
    "msg": "Error executing command.",
    "rc": 2,
    "stderr": "",
    "stderr_lines": [],
    "stdout": "",
    "stdout_lines": []
}
[WARNING]: Host 'server1' is using the discovered Python interpreter at '/usr/bin/python3.12', but future installation o
f another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-
core/2.21/reference_appendices/interpreter_discovery.html for more information.
```

```
[ERROR]: Task failed: Module failed: Failed to lock apt for exclusive operation: Failed to lock directory /var/lib/apt/l
ists/: E:Could not open lock file /var/lib/apt/lists/lock - open (13: Permission denied)
Origin: <adhoc 'apt' task>

{'action': 'apt', 'args': {'update_cache': 'true'}, 'timeout': 0, 'async_val': 0, 'poll': 15}

server1 | FAILED! => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.12"
    },
    "changed": false,
    "msg": "Failed to lock apt for exclusive operation: Failed to lock directory /var/lib/apt/lists/: E:Could not open l
ock file /var/lib/apt/lists/lock - open (13: Permission denied)"
}
[WARNING]: Host 'server2' is using the discovered Python interpreter at '/usr/bin/python3.12', but future installation o
f another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-
core/2.21/reference_appendices/interpreter_discovery.html for more information.
server2 | FAILED! => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.12"
    },
    "changed": false,
    "msg": "Failed to lock apt for exclusive operation: Failed to lock directory /var/lib/apt/lists/: E:Could not open l
ock file /var/lib/apt/lists/lock - open (13: Permission denied)"
}
```

Try editing the command and add something that would elevate the privilege. Issue the command ansible all -m apt -a update_cache=true --become --ask-become-pass. Enter the sudo password when prompted. You will notice now that the output of this command is a success. The update_cache=true is the same thing as running sudo apt update. The --become command elevates the privileges and the --ask-become-pass asks for the password. For now, even if we only changed the packaged index, we were able to change something on the remote server. You may notice after the second command was executed, the status is

CHANGED compared to the first command, which is FAILED

```
charles-vence-espinas@workstation:~/ansible-lab$ ansible all -m apt -a update_cache=true --become --ask-become-pass
BECOME password:
[WARNING]: Host 'centOS' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of
another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-co
re/2.21/reference_appendices/interpreter_discovery.html for more information.
[WARNING]: Updating cache and auto-installing missing dependency: python3-apt
[ERROR]: Task failed: Module failed: Error executing command: [Errno 2] No such file or directory: b'update'
Origin: <adhoc 'apt' task>

{'action': 'apt', 'args': {'update_cache': 'true'}, 'timeout': 0, 'async_val': 0, 'poll': 15}

centOS | FAILED! => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.9"
    },
    "changed": false,
    "cmd": "update",
    "msg": "Error executing command.",
    "rc": 2,
    "stderr": "",
    "stderr_lines": [],
    "stdout": "",
    "stdout_lines": []
}
[WARNING]: Host 'server2' is using the discovered Python interpreter at '/usr/bin/python3.12', but future installation o
f another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-
core/2.21/reference_appendices/interpreter_discovery.html for more information.
```

```
server2 | CHANGED => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.12"
    },
    "cache_update_time": 1787240830,
    "cache_updated": true,
    "changed": true
}
[WARNING]: Host 'server1' is using the discovered Python interpreter at '/usr/bin/python3.12', but future installation o
f another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-
core/2.21/reference_appendices/interpreter_discovery.html for more information.
server1 | CHANGED => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.12"
    },
    "cache_update_time": 1787240830,
    "cache_updated": true,
    "changed": true
}
```

For Cent OS

```
charles-vence-espinas@workstation:~/CPE212_CharlesVenceEspinas/ansible-lab$ ansible centos -m dnf -a "update_cache=true"
 --become --ask-become-pass
BECOME password:
[WARNING]: Host 'centOS' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of
another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-co
re/2.21/reference_appendices/interpreter_discovery.html for more information.
centOS | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.9"
    },
    "changed": false,
    "msg": "Cache updated",
    "rc": 0,
    "results": []
}
```

2. Let's try to install VIM, which is an almost compatible version of the UNIX editor Vi. To do this, we will just changed the module part in 1.1 instruction. Here is the command: ansible all -m apt -a name=vim-nox --become --ask-become-pass. The command would take some time after typing the password because the local machine instructed the remote servers to actually install the package.

```
charles-vence-espinas@workstation:~/ansible-lab$ ansible all -m apt -a name=vim-nox --become --ask-become-pass
BECOME password:
[WARNING]: Host 'centOS' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of
another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-co
re/2.21/reference_appendices/interpreter_discovery.html for more information.
[WARNING]: Updating cache and auto-installing missing dependency: python3-apt
[ERROR]: Task failed: Module failed: Error executing command: [Errno 2] No such file or directory: b'update'
Origin: <adhoc 'apt' task>

{'action': 'apt', 'args': {'name': 'vim-nox'}, 'timeout': 0, 'async_val': 0, 'poll': 15}

centOS | FAILED! => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.9"
    },
    "changed": false,
    "cmd": "update",
    "msg": "Error executing command.",
    "rc": 2,
    "stderr": "",
    "stderr_lines": [],
    "stdout": "",
    "stdout_lines": []
}
[WARNING]: Host 'server2' is using the discovered Python interpreter at '/usr/bin/python3.12', but future installation o
f another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-
core/2.21/reference_appendices/interpreter_discovery.html for more information.
```

```
server2 | CHANGED => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.12"
    },
    "cache_update_time": 1787240830,
    "cache_updated": false,
    "changed": true,
    "stderr": "",
    "stderr_lines": [],
    "stdout": "Reading package lists...\nBuilding dependency tree...\nReading state information...\nThe following packa
es were automatically installed and are no longer required:\n  libfwupd2 libwoff1\nUse 'sudo apt autoremove' to remove
hem.\nThe following additional packages will be installed:\n  fonts-lato javascript-common libjs-jquery liblua5.1-0 lib
uby libruby3.2\n  rake ruby ruby-net-telnet ruby-rubygems ruby-sdbm ruby-webrick ruby-xmlrpc\n  ruby3.2 rubygems-integra
tion vim-runtime\nSuggested packages:\n  apache2 | lighttpd | httpd ri ruby-dev bundler cscope vim-doc\nThe following N
W packages will be installed:\n  fonts-lato javascript-common libjs-jquery liblua5.1-0 libruby libruby3.2\n  rake ruby
uby-net-telnet ruby-rubygems ruby-sdbm ruby-webrick ruby-xmlrpc\n  ruby3.2 rubygems-integration vim-nox vim-runtime\n0
pgraded, 17 newly installed, 0 to remove and 62 not upgraded.\nNeed to get 18.4 MB of archives.\nAfter this operation,
3.8 MB of additional disk space will be used.\nGet:1 http://ph.archive.ubuntu.com/ubuntu noble/main amd64 fonts-lato al
 2.015-1 [2781 kB]\nGet:2 http://ph.archive.ubuntu.com/ubuntu noble/main amd64 javascript-common all 11+nmu1 [5936 B]\n
et:3 http://ph.archive.ubuntu.com/ubuntu noble/main amd64 libjs-jquery all 3.6.1+dfsg+~3.5.14-1 [328 kB]\nGet:4 http://
h.archive.ubuntu.com/ubuntu noble/universe amd64 liblua5.1-0 amd64 5.1.5-9build2 [120 kB]\nGet:5 http://ph.archive.ubunt
u.com/ubuntu noble/main amd64 rubygems-integration all 1.18 [5336 B]\nGet:6 http://ph.archive.ubuntu.com/ubuntu noble-u
dates/main amd64 ruby3.2 amd64 3.2.3-1ubuntu0.24.04.8 [50.7 kB]\nGet:7 http://ph.archive.ubuntu.com/ubuntu noble/main a
d64 ruby-rubygems all 3.4.20-1 [238 kB]\nGet:8 http://ph.archive.ubuntu.com/ubuntu noble/main amd64 ruby amd64 1:3.2~ub
ntu1 [3466 B]\nGet:9 http://ph.archive.ubuntu.com/ubuntu noble/main amd64 rake all 13.0.6-3 [61.6 kB]\nGet:10 http://ph
archive.ubuntu.com/ubuntu noble/main amd64 ruby-net-telnet all 0.2.0-1 [13.3 kB]\nGet:11 http://ph.archive.ubuntu.com/u
```

```
[WARNING]: Host 'server1' is using the discovered Python interpreter at '/usr/bin/python3.12', but future installation o
f another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-
core/2.21/reference_appendices/interpreter_discovery.html for more information.
server1 | CHANGED => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.12"
    },
    "cache_update_time": 1787240830,
    "cache_updated": false,
    "changed": true,
    "stderr": "",
    "stderr_lines": [],
    "stdout": "Reading package lists...\nBuilding dependency tree...\nReading state information...\nThe following packag
es were automatically installed and are no longer required:\n  libfwupd2 libwoff1\nUse 'sudo apt autoremove' to remove t
hem.\nThe following additional packages will be installed:\n  fonts-lato javascript-common libjs-jquery liblua5.1-0 libr
uby libruby3.2\n  rake ruby ruby-net-telnet ruby-rubygems ruby-sdbm ruby-webrick ruby-xmlrpc\n  ruby3.2 rubygems-integra
tion vim-runtime\nSuggested packages:\n  apache2 | lighttpd | httpd ri ruby-dev bundler cscope vim-doc\nThe following NE
W packages will be installed:\n  fonts-lato javascript-common libjs-jquery liblua5.1-0 libruby libruby3.2\n  rake ruby r
uby-net-telnet ruby-rubygems ruby-sdbm ruby-webrick ruby-xmlrpc\n  ruby3.2 rubygems-integration vim-nox vim-runtime\n0 u
pgraded, 17 newly installed, 0 to remove and 62 not upgraded.\nNeed to get 18.4 MB of archives.\nAfter this operation, 8
3.8 MB of additional disk space will be used.\nGet:1 http://ph.archive.ubuntu.com/ubuntu noble/main amd64 fonts-lato all
 2.015-1 [2781 kB]\nGet:2 http://ph.archive.ubuntu.com/ubuntu noble/main amd64 javascript-common all 11+nmu1 [5936 B]\nG
et:3 http://ph.archive.ubuntu.com/ubuntu noble/main amd64 libjs-jquery all 3.6.1+dfsg+~3.5.14-1 [328 kB]\nGet:4 http://p
h.archive.ubuntu.com/ubuntu noble/universe amd64 liblua5.1-0 amd64 5.1.5-9build2 [120 kB]\nGet:5 http://ph.archive.ubunt
u.com/ubuntu noble/main amd64 rubygems-integration all 1.18 [5336 B]\nGet:6 http://ph.archive.ubuntu.com/ubuntu noble-up
dates/main amd64 ruby3.2 amd64 3.2.3-1ubuntu0.24.04.8 [50.7 kB]\nGet:7 http://ph.archive.ubuntu.com/ubuntu noble/main am
d64 ruby-rubygems all 3.4.20-1 [238 kB]\nGet:8 http://ph.archive.ubuntu.com/ubuntu noble/main amd64 ruby amd64 1:3.2~ubu
ntu1 [3466 B]\nGet:9 http://ph.archive.ubuntu.com/ubuntu noble/main amd64 rake all 13.0.6-3 [61.6 kB]\nGet:10 http://ph.
archive.ubuntu.com/ubuntu noble/main amd64 ruby-net-telnet all 0.2.0-1 [13.3 kB]\nGet:11 http://ph.archive.ubuntu.com/ub
```

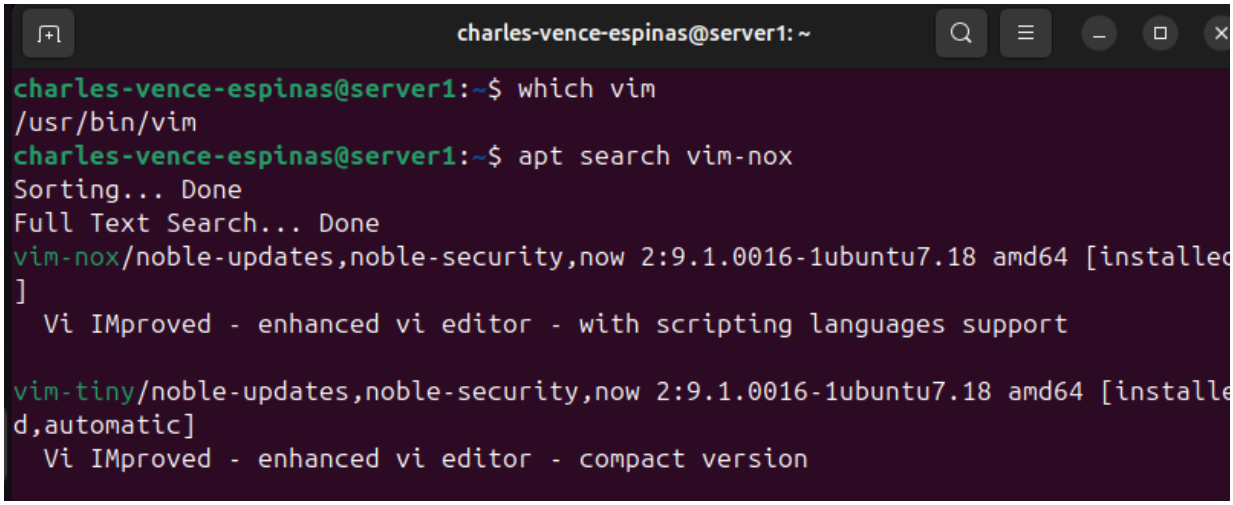
For CentOS

```
charles-vence-espinas@workstation:~/CPE212_CharlesVenceEspinas/ansible-lab$ ansible centos -m dnf -a "name=vim-enhanced"
 --become --ask-become-pass
BECOME password:
[WARNING]: Host 'centOS' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of
another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-co
re/2.21/reference_appendices/interpreter_discovery.html for more information.
centOS | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.9"
    },
    "changed": false,
    "msg": "Nothing to do",
    "rc": 0,
    "results": []
}
```

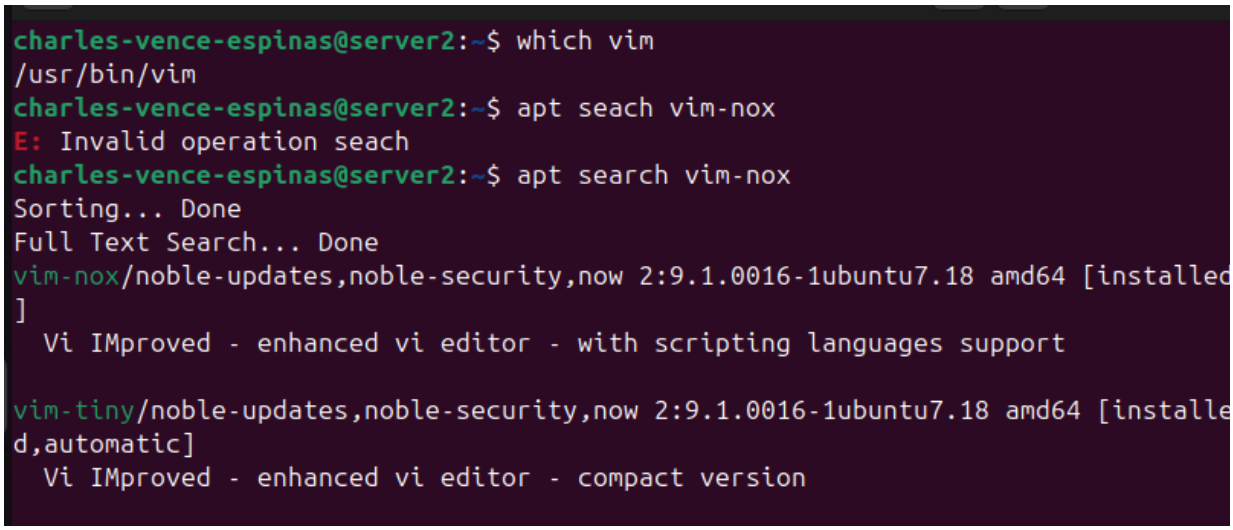
a. Verify that you have installed the package in the remote servers. Issue the command which vim and the command apt search vim-nox respectively. Was the command successful?

- Yes, which vim showed the path /usr/bin/vim which confirm that Vim was installed. apt search vim-nox confirmed vim-nox was installed on ubuntu servers while dnf info vim-enhanced confirmed the package on the centos

Server1



Server2



CentOS

```
[charles-vence-espinas@centOS ~]$ dnf info vim-enhanced
CentOS Stream 9 - BaseOS                        4.1 MB/s | 9.0 MB     00:02
CentOS Stream 9 - AppStream                     7.8 MB/s |  29 MB     00:03
CentOS Stream 9 - Extras packages                30 kB/s |  22 kB     00:00
Installed Packages
Name         : vim-enhanced
Epoch        : 2
Version      : 8.2.2637
Release      : 33.el9
Architecture : x86_64
Size         : 3.8 M
Source       : vim-8.2.2637-33.el9.src.rpm
Repository   : @System
From repo    : AppStream
Summary      : A version of the VIM editor which includes recent enhancements
URL          : http://www.vim.org/
License      : Vim and MIT
Description  : VIM (VIsual editor iMproved) is an updated and improved version
             : of the vi editor.  Vi was the first real screen-based editor for
             : UNIX, and is still very popular.  VIM improves on vi by adding
             : new features: multiple windows, multi-level undo, block
             : highlighting and more.  The vim-enhanced package contains a
             : version of VIM with extra, recently introduced features like
             : Python and Perl interpreters.
             :
             : Install the vim-enhanced package if you'd like to use a version
             : of the VIM editor which includes recently added enhancements like
             : interpreters for the Python and Perl scripting languages.  You'll
             : also need to install the vim-common package.
```

```
Available Packages
Name         : vim-enhanced
Epoch        : 2
Version      : 8.2.2637
Release      : 40.el9
Architecture : x86_64
Size         : 1.7 M
Source       : vim-8.2.2637-40.el9.src.rpm
Repository   : appstream
Summary      : A version of the VIM editor which includes recent enhancements
URL          : http://www.vim.org/
License      : Vim and MIT
Description  : VIM (VIsual editor iMproved) is an updated and improved version
             : of the vi editor.  Vi was the first real screen-based editor for
             : UNIX, and is still very popular.  VIM improves on vi by adding
             : new features: multiple windows, multi-level undo, block
             : highlighting and more.  The vim-enhanced package contains a
             : version of VIM with extra, recently introduced features like
             : Python and Perl interpreters.
```

b. Check the logs in the servers using the following commands: cd /var/log. After this, issue the command ls, go to the folder apt and open history.log. Describe what you see in the history.log.

- It contains the records of package activities including the date and time, commands used, users and packages that were installed, updated or configured.

```
charles-vence-espinas@workstation:~/ansible-lab$ cd /var/log
charles-vence-espinas@workstation:/var/log$ ls
alternatives.log    boot.log.4             dmesg.0          hp               sssd
alternatives.log.1  boot.log.5             dmesg.1.gz       installer        syslog
apport.log          bootstrap.log          dmesg.2.gz       journal          syslog.1
apt                 btmp                   dmesg.3.gz       kern.log         syslog.2.gz
auth.log            btmp.1                 dmesg.4.gz       kern.log.1       sysstat
auth.log.1          cloud-init.log         dpkg.log         kern.log.2.gz    ubuntu-advantage-apt-hook.log
auth.log.2.gz       cloud-init-output.log  dpkg.log.1       lastlog          ufw.log
boot.log            cups                   faillog          openvpn          ufw.log.1
boot.log.1          cups-browsed           fontconfig.log   private          ufw.log.2.gz
boot.log.2          dist-upgrade           gdm3             README           unattended-upgrades
boot.log.3          dmesg                  gpu-manager.log  speech-dispatcher  wtmp
charles-vence-espinas@workstation:/var/log$ cd apt
charles-vence-espinas@workstation:/var/log/apt$ ls
eipp.log.xz  history.log  history.log.1.gz  term.log  term.log.1.gz
```

```
 GNU nano 7.2                                    history.log

Start-Date: 2026-08-19  00:12:03
Commandline: apt install python3-pip -y
Requested-By: charles-vence-espinas (1000)
Install: libgcc-13-dev:amd64 (13.3.0-6ubuntu2~24.04.1, automatic), libstdc++-13-dev:amd64 (13.3.0-6ubuntu2~24.04.1, aut>
End-Date: 2026-08-19  00:12:23

Start-Date: 2026-08-19  00:12:46
Commandline: apt install ansible
Requested-By: charles-vence-espinas (1000)
Upgrade: ansible:amd64 (14.2.0-1ppa~noble, 14.3.1-1ppa~noble)
End-Date: 2026-08-19  00:13:57

Start-Date: 2026-08-20  23:48:22
Commandline: /usr/bin/unattended-upgrade
Upgrade: libfreerdp3-3:amd64 (3.30.0+dfsg-0ubuntu0.24.04.1, 3.30.0+dfsg-0ubuntu0.24.04.2), libwinpr3-3:amd64 (3.30.0+df>
End-Date: 2026-08-20  23:48:25

Start-Date: 2026-08-20  23:48:31
Commandline: /usr/bin/unattended-upgrade
Upgrade: udev:amd64 (255.4-1ubuntu8.16, 255.4-1ubuntu8.17), systemd-oomd:amd64 (255.4-1ubuntu8.16, 255.4-1ubuntu8.17), >
End-Date: 2026-08-20  23:49:01

Start-Date: 2026-08-20  23:49:06
Commandline: /usr/bin/unattended-upgrade
Upgrade: libssl3t64:amd64 (3.0.13-0ubuntu3.11, 3.0.13-0ubuntu3.12), openssl:amd64 (3.0.13-0ubuntu3.11, 3.0.13-0ubuntu3.>
End-Date: 2026-08-20  23:49:09
```

3. This time, we will install a package called snapd. Snap is pre-installed in Ubuntu system. However, our goal is to create a command that checks for the latest installation package.

a. Issue the command: ansible all -m apt -a name=snapd --become --ask-become-pass
Can you describe the result of this command? Is it a success? Did it change anything in the remote servers?

- Yes, it succeeded on server 1 and server 2 and it returned success. It failed first on centos but after I used an dnf module, it succeeded.

```
charles-vence-espinas@workstation:~/ansible-lab$ ansible all -m apt -a name=snapd --become --ask-become-pass
BECOME password:
[WARNING]: Host 'centOS' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of
another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-co
re/2.21/reference_appendices/interpreter_discovery.html for more information.
[WARNING]: Updating cache and auto-installing missing dependency: python3-apt
[ERROR]: Task failed: Module failed: Error executing command: [Errno 2] No such file or directory: b'update'
Origin: <adhoc 'apt' task>

{'action': 'apt', 'args': {'name': 'snapd'}, 'timeout': 0, 'async_val': 0, 'poll': 15}

centOS | FAILED! => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.9"
    },
    "changed": false,
    "cmd": "update",
    "msg": "Error executing command.",
    "rc": 2,
    "stderr": "",
    "stderr_lines": [],
    "stdout": "",
    "stdout_lines": []
}
```

```
[WARNING]: Host 'server2' is using the discovered Python interpreter at '/usr/bin/python3.12', but future installation o
f another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-
core/2.21/reference_appendices/interpreter_discovery.html for more information.
server2 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.12"
    },
    "cache_update_time": 1787240830,
    "cache_updated": false,
    "changed": false
}
[WARNING]: Host 'server1' is using the discovered Python interpreter at '/usr/bin/python3.12', but future installation o
f another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-
core/2.21/reference_appendices/interpreter_discovery.html for more information.
server1 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.12"
    },
    "cache_update_time": 1787240830,
    "cache_updated": false,
    "changed": false
}
```

For CentOS

```
charles-vence-espinas@workstation:~/CPE212_CharlesVenceEspinas/ansible-lab$ ansible centos -m dnf -a "name=curl" --becom
e --ask-become-pass
BECOME password:
[WARNING]: Host 'centOS' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of
another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-co
re/2.21/reference_appendices/interpreter_discovery.html for more information.
centOS | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.9"
    },
    "changed": false,
    "msg": "Nothing to do",
    "rc": 0,
    "results": []
}
```

b. Now, try to issue this command: ansible all -m apt -a "name=snapd state=latest" --become --ask-become-pass

Describe the output of this command. Notice how we added the command state=latest and placed them in double quotations.

- Both server 1 and server 2 returned success and also centos returned success after using dnf module.

```
charles-vence-espinas@workstation:~/ansible-lab$ ansible all -m apt -a "name=snapd state=latest" --become --ask-become-p
ass
BECOME password:
[WARNING]: Host 'centOS' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of
another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-co
re/2.21/reference_appendices/interpreter_discovery.html for more information.
[WARNING]: Updating cache and auto-installing missing dependency: python3-apt
[ERROR]: Task failed: Module failed: Error executing command: [Errno 2] No such file or directory: b'update'
Origin: <adhoc 'apt' task>

{'action': 'apt', 'args': {'name': 'snapd', 'state': 'latest'}, 'timeout': 0, 'async_val': 0, 'poll': 15}

centOS | FAILED! => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.9"
    },
    "changed": false,
    "cmd": "update",
    "msg": "Error executing command.",
    "rc": 2,
    "stderr": "",
    "stderr_lines": [],
    "stdout": "",
    "stdout_lines": []
}
```

```
[WARNING]: Host 'server2' is using the discovered Python interpreter at '/usr/bin/python3.12', but future installation o
f another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-
core/2.21/reference_appendices/interpreter_discovery.html for more information.
server2 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.12"
    },
    "cache_update_time": 1787240830,
    "cache_updated": false,
    "changed": false
}
[WARNING]: Host 'server1' is using the discovered Python interpreter at '/usr/bin/python3.12', but future installation o
f another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-
core/2.21/reference_appendices/interpreter_discovery.html for more information.
server1 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.12"
    },
    "cache_update_time": 1787240830,
    "cache_updated": false,
    "changed": false
}
```

For CentOS

```
charles-vence-espinas@workstation:~/CPE212_CharlesVenceEspinas/ansible-lab$ ansible centos -m dnf -a "name=curl state=la
test" --become --ask-become-pass
BECOME password:
[WARNING]: Host 'centOS' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of
another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-co
re/2.21/reference_appendices/interpreter_discovery.html for more information.
centOS | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.9"
    },
    "changed": false,
    "msg": "Nothing to do",
    "rc": 0,
    "results": []
}
```

Task 2: Writing our First Playbook

1. With ad hoc commands, we can simplify the administration of remote servers. For example, we can install

updates, packages, and applications, etc. However, the real strength of ansible comes from its playbooks. When we write a playbook, we can define the state that we want our servers to be in and the place or commands that ansible will carry out to bring to that state. You can use an editor to create a playbook. Before we proceed, make sure that you are in the directory of the repository that we use in the previous activities (CPE232_yourname). Issue the command nano install_apache.yml. This will create a playbook file called install_apache.yaml. The .yaml is the basic standard extension for playbook files.
When the editor appears, type the following:
Make sure to save the file. Take note also of the alignments of the texts. Use TAB instead of SPACE BAR for spacing.

```
 GNU nano 7.2                                    install_apache.yml
---
- hosts: all
  become: true
  tasks:

    - name: install apache2 package
      apt:
        name: apache2
```

With CentOS

```
 GNU nano 7.2                                    install_apache.yml *
---
- hosts: ubuntu
  become: true
  tasks:

    - name: install apache2 package
      apt:
        name: apache2

- hosts: centos
  become: true
  tasks:
    - name: install httpd package
      dnf:
        name: httpd

    - name: start httpd service
      service:
        name: httpd
        state: started
        enabled: true
```

2. Run the yml file using the command: ansible-playbook --ask-become-pass install_apache.yml. **Describe the result of this command.**

- The first run failed on the centos because it used apt. After adding separate tasks for ubuntu and centos, the paybook successfully installed and started apache on all servers.

```
charles-vence-espinas@workstation:~/CPE212_CharlesVenceEspinas/ansible-lab$ ansible-playbook --ask-become-pass install_a
pache.yml
BECOME password:

PLAY [all] ********************************************************************************************************

TASK [Gathering Facts] ********************************************************************************************
[WARNING]: Host 'server1' is using the discovered Python interpreter at '/usr/bin/python3.12', but future installation o
f another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-
core/2.21/reference_appendices/interpreter_discovery.html for more information.
ok: [server1]
[WARNING]: Host 'server2' is using the discovered Python interpreter at '/usr/bin/python3.12', but future installation o
f another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-
core/2.21/reference_appendices/interpreter_discovery.html for more information.
ok: [server2]
[WARNING]: Host 'centOS' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of
another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-co
re/2.21/reference_appendices/interpreter_discovery.html for more information.
ok: [centOS]

TASK [install apache2 package] ************************************************************************************
[WARNING]: Updating cache and auto-installing missing dependency: python3-apt
[ERROR]: Task failed: Module failed: Error executing command: [Errno 2] No such file or directory: b'update'
Origin: /home/charles-vence-espinas/CPE212_CharlesVenceEspinas/ansible-lab/install_apache.yml:6:7

4   tasks:
5
6     - name: install apache2 package
        ^ column 7

fatal: [centOS]: FAILED! => {"changed": false, "cmd": "update", "msg": "Error executing command.", "rc": 2, "stderr": ""
, "stderr_lines": [], "stdout": "", "stdout_lines": []}
changed: [server1]
changed: [server2]

PLAY RECAP ********************************************************************************************************
centOS                     : ok=1    changed=0    unreachable=0    failed=1    skipped=0    rescued=0    ignored=0
server1                    : ok=2    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
server2                    : ok=2    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

With CentOS

```
charles-vence-espinas@workstation:~/CPE212_CharlesVenceEspinas/ansible-lab$ ansible-playbook --ask-become-pass install_a
pache.yml
BECOME password:

PLAY [ubuntu] *****************************************************************************************************

TASK [Gathering Facts] ********************************************************************************************
[WARNING]: Host 'server1' is using the discovered Python interpreter at '/usr/bin/python3.12', but future installation o
f another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-
core/2.21/reference_appendices/interpreter_discovery.html for more information.
ok: [server1]
[WARNING]: Host 'server2' is using the discovered Python interpreter at '/usr/bin/python3.12', but future installation o
f another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-
core/2.21/reference_appendices/interpreter_discovery.html for more information.
ok: [server2]

TASK [install apache2 package] ************************************************************************************
ok: [server2]
ok: [server1]

PLAY [centos] *****************************************************************************************************

TASK [Gathering Facts] ********************************************************************************************
[WARNING]: Host 'centOS' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of
another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-co
re/2.21/reference_appendices/interpreter_discovery.html for more information.
ok: [centOS]
```

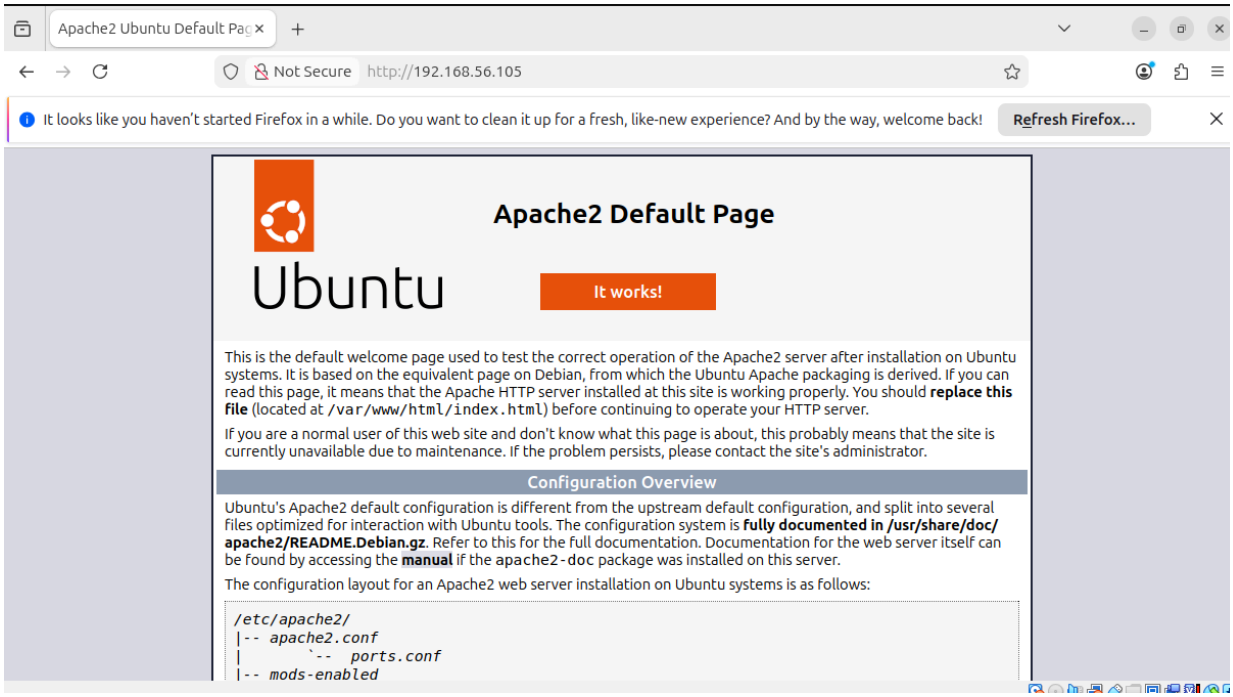
```
TASK [install httpd package] ***************************************************************************************
changed: [centOS]

TASK [start httpd service] *****************************************************************************************
changed: [centOS]

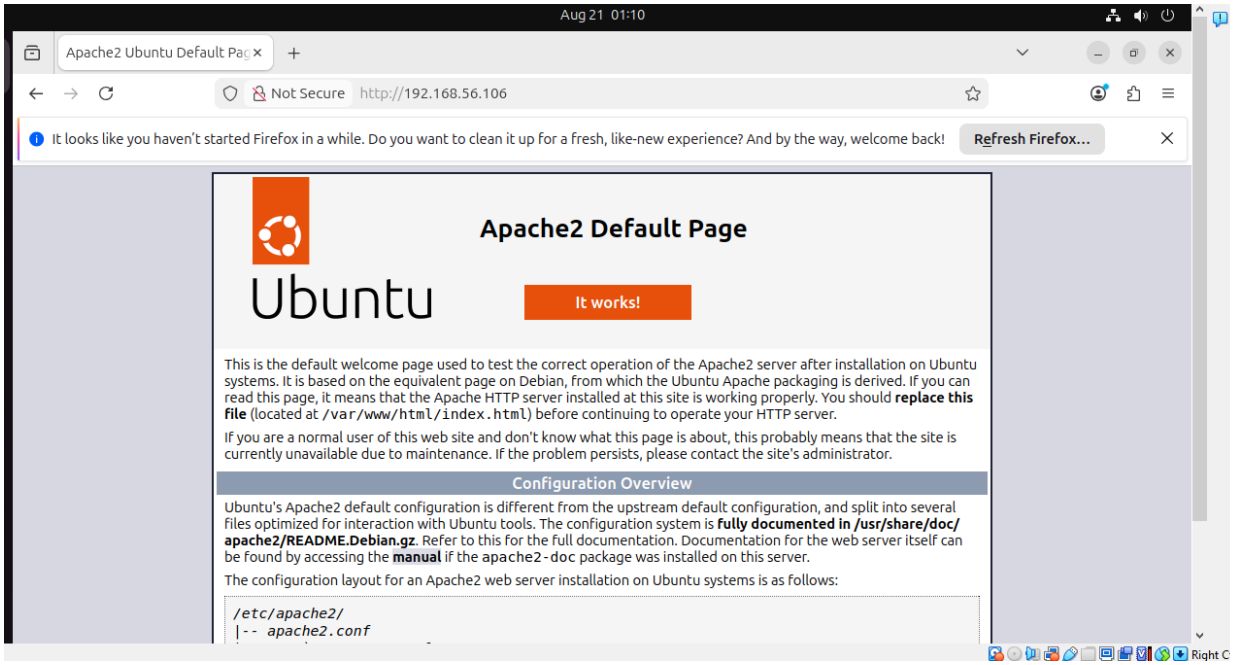
PLAY RECAP *********************************************************************************************************
centOS                     : ok=3    changed=2    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
server1                    : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
server2                    : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

3. To verify that apache2 was installed automatically in the remote servers, go to the web browsers on each server and type its IP address. You should see something like this.

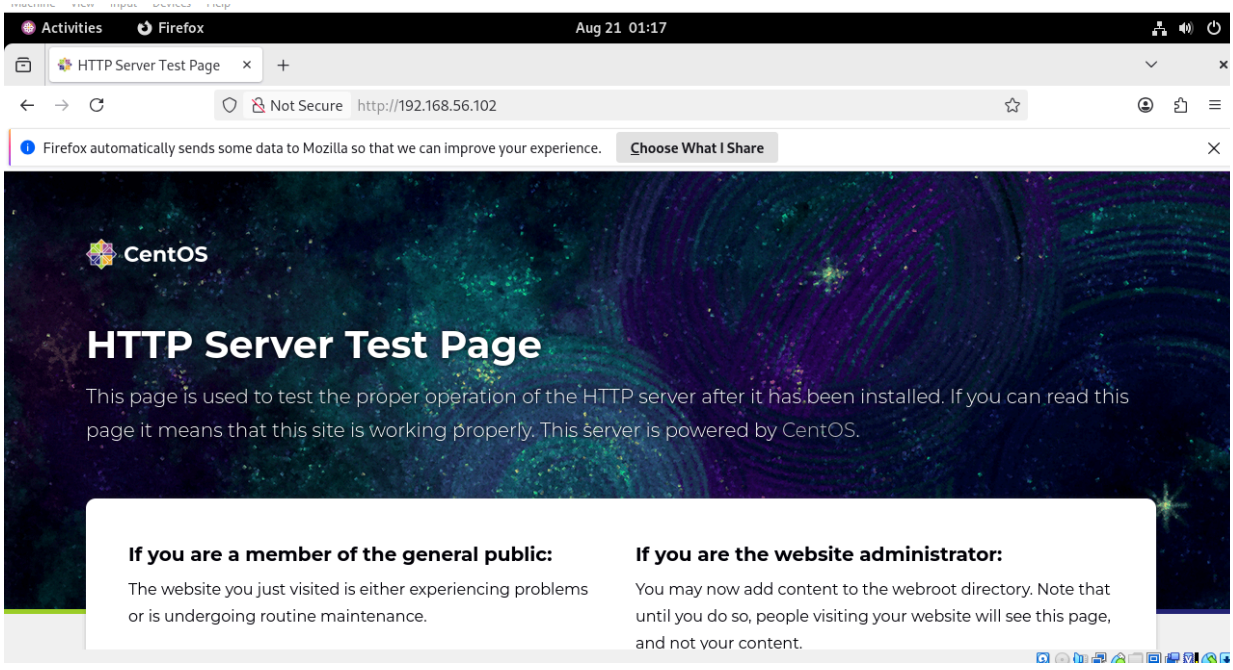
Server1



Server 2



CentOS



4. **Try to edit the install_apache.yml and change the name of the package to any name that will not be recognized. What is the output?**

- The playbook returned failed because the package manager could not find the invalid package name in the repositories.

```
  GNU nano 7.2                                        install_apache.yml
---
- hosts: ubuntu
  become: true
  tasks:

    - name: install apache2 package
      apt:
        name: apache2-invalid

- hosts: centos
  become: true
  tasks:
    - name: install httpd package
      dnf:
        name: httpd-invalid

    - name: start httpd service
      service:
        name: httpd
        state: started
        enabled: true
```

Output

```
charles-vence-espinas@workstation:~/CPE212_CharlesVenceEspinas/ansible-lab$ ansible-playbook --ask-become-pass install_a
pache.yml
BECOME password:

PLAY [ubuntu] ***************************************************************************************************

TASK [Gathering Facts] ******************************************************************************************
[WARNING]: Host 'server2' is using the discovered Python interpreter at '/usr/bin/python3.12', but future installation o
f another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-
core/2.21/reference_appendices/interpreter_discovery.html for more information.
ok: [server2]
[WARNING]: Host 'server1' is using the discovered Python interpreter at '/usr/bin/python3.12', but future installation o
f another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-
core/2.21/reference_appendices/interpreter_discovery.html for more information.
ok: [server1]

TASK [install apache2 package] **********************************************************************************
[ERROR]: Task failed: Module failed: No package matching 'apache2-invalid' is available
Origin: /home/charles-vence-espinas/CPE212_CharlesVenceEspinas/ansible-lab/install_apache.yml:6:7

4   tasks:
5
6     - name: install apache2 package
        ^ column 7

fatal: [server2]: FAILED! => {"changed": false, "msg": "No package matching 'apache2-invalid' is available"}
fatal: [server1]: FAILED! => {"changed": false, "msg": "No package matching 'apache2-invalid' is available"}

PLAY RECAP ******************************************************************************************************
server1                    : ok=1    changed=0    unreachable=0    failed=1    skipped=0    rescued=0    ignored=0
server2                    : ok=1    changed=0    unreachable=0    failed=1    skipped=0    rescued=0    ignored=0
```

5. This time, we are going to put additional tasks into our playbook. Edit the install_apache.yml. As you can see, we are now adding an additional command, which is the update_cache. This command updates existing package-indexes on a supporting distro but not upgrading installed-packages (utilities) that were being installed.
Save the changes to this file and exit.

```
---
- hosts: ubuntu
  become: true
  tasks:

    - name: update repository index
      apt:
        update_cache: yes

    - name: install apache2 package
      apt:
        name: apache2

- hosts: centos
  become: true
  tasks:

    - name: update repository index
      dnf:
        update_cache: yes

    - name: install httpd package
      dnf:
        name: httpd
```

```
    - name: start httpd service
      service:
        name: httpd
        state: started
        enabled: true
```

6. Run the playbook and describe the output. Did the new command change anything on the remote servers?

- Yes, the update_cache : yes task refreshed the APT package lists on the server 1 and server 2 so the y showed changes. On centos, it showed ok because its DNF cache was already up to date. Apache showed ok because it was already installed.

```
charles-vence-espinas@workstation:~/CPE212_CharlesVenceEspinas/ansible-lab$ ansible-playbook --ask-become-pass install_a
pache.yml
BECOME password:

PLAY [ubuntu] ***********************************************************************************************************

TASK [Gathering Facts] **************************************************************************************************
[WARNING]: Host 'server1' is using the discovered Python interpreter at '/usr/bin/python3.12', but future installation c
f another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-
core/2.21/reference_appendices/interpreter_discovery.html for more information.
ok: [server1]
[WARNING]: Host 'server2' is using the discovered Python interpreter at '/usr/bin/python3.12', but future installation c
f another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-
core/2.21/reference_appendices/interpreter_discovery.html for more information.
ok: [server2]

TASK [update repository index] ******************************************************************************************
changed: [server1]
changed: [server2]

TASK [install apache2 package] ******************************************************************************************
ok: [server1]
ok: [server2]
```

```
PLAY [centos] ***********************************************************************************************************

TASK [Gathering Facts] **************************************************************************************************
[WARNING]: Host 'centOS' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of
another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-co
re/2.21/reference_appendices/interpreter_discovery.html for more information.
ok: [centOS]

TASK [update repository index] ******************************************************************************************
ok: [centOS]

TASK [install httpd package] ********************************************************************************************
ok: [centOS]

TASK [start httpd service] **********************************************************************************************
ok: [centOS]

PLAY RECAP **************************************************************************************************************
centOS                     : ok=4    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
server1                    : ok=3    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
server2                    : ok=3    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

7. Edit again the install_apache.yml. This time, we are going to add a PHP support for the apache package we installed earlier.
Save the changes to this file and exit.

```yaml
---
- hosts: ubuntu
  become: true
  tasks:

    - name: update repository index
      apt:
        update_cache: yes

    - name: install apache2 package
      apt:
        name: apache2

    - name: add PHP support for apache
      apt:
        name: libapache2-mod-php

- hosts: centos
  become: true
  tasks:

    - name: update repository index
      dnf:
        update_cache: yes
```

```yaml
    - name: install httpd package
      dnf:
        name: httpd

    - name: add PHP support for apache
      dnf:
        name: php
        state: present

    - name: start httpd service
      service:
        name: httpd
        state: started
        enabled: true
```

8. Run the playbook and describe the output. Did the new command change anything on the remote servers?

- Yes, the task installed PHP support with libapache2-mod-php on ubuntu and php on centos so all the servers showed changed.

```
charles-vence-espinas@workstation:~/CPE212_CharlesVenceEspinas/ansible-lab$ ansible-playbook --ask-become-pass install_a
pache.yml
BECOME password:

PLAY [ubuntu] ******************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************
[WARNING]: Host 'server1' is using the discovered Python interpreter at '/usr/bin/python3.12', but future installation o
f another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-
core/2.21/reference_appendices/interpreter_discovery.html for more information.
ok: [server1]
[WARNING]: Host 'server2' is using the discovered Python interpreter at '/usr/bin/python3.12', but future installation o
f another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-
core/2.21/reference_appendices/interpreter_discovery.html for more information.
ok: [server2]

TASK [update repository index] *************************************************************************************
changed: [server2]
changed: [server1]

TASK [install apache2 package] *************************************************************************************
ok: [server1]
ok: [server2]
```

```
TASK [add PHP support for apache] **********************************************************************************
changed: [server1]
changed: [server2]

PLAY [centos] ******************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************
[WARNING]: Host 'centOS' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of
another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-co
re/2.21/reference_appendices/interpreter_discovery.html for more information.
ok: [centOS]

TASK [update repository index] *************************************************************************************
ok: [centOS]

TASK [install httpd package] ***************************************************************************************
ok: [centOS]

TASK [add PHP support for apache] **********************************************************************************
changed: [centOS]

TASK [start httpd service] *****************************************************************************************
ok: [centOS]
```

```
PLAY RECAP *********************************************************************************************************
centOS                     : ok=5    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
server1                    : ok=4    changed=2    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
server2                    : ok=4    changed=2    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

9. Finally, make sure that we are in sync with GitHub. Provide the link of your GitHub repository
https://github.com/charlesespinas/CPE212_CharlesVenceEspinas/tree/main/ansible-lab

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Answer the following:

1. What is the importance of using a playbook?

- An ansible playbook is important because it allows us to automate many tasks at once. It makes the tasks easier to repeat and manage on many servers. It also helps us to reduce mistakes because we do not

have to enter every command manually. Playbooks can also be used for different operating systems by using the proper modules and tasks.

2. Summarize what we have done on this activity.

Conclusions/Learnings:

- In this activity, I have learned how to correctly use Ansible ad hoc commands to manage remote servers. We used –become to get the administrator privileges and installed checked packages and servers. We also created and modified the install_apahe.yml playbook to work with both Ubuntu and CentOS servers. We used these playbooks to update the packages lists, install and start apache and add PHP support. Lastly, we checked the web server using the ip address of each server in the browser to make sure that it was working.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

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